UPI: RHYTHM OF MONEY

Decoding India's Payment Behaviour | 2016 – 2025

An End-to-End SQL + Power BI Analysis of NPCI Open Data

Metric	Value
Total Transactions (2016–2026)	706,864 million
Period Covered	August 2016 – February 2026
Banks on Network (Aug 2016 → Feb 2026)	21 → 694
Average Ticket Size (FY16 → FY25)	₹2,648 → ₹1,310
Peak Monthly Volume	21,703 Mn (January 2026)
Data Source	National Payments Corporation of India (NPCI)
Analysis Tools	SQL Server 2019 · Power BI · Excel Power Query

Executive Summary

This report presents the findings of a comprehensive analysis of India's Unified Payments Interface (UPI) spanning nearly a decade — from its quiet launch in April 2016 to its position as the world's largest real-time payments network by 2025. Using NPCI's publicly available data and 13 purpose-built SQL queries across five analytical sections, this study moves beyond headline numbers to reveal the behavioural, seasonal, and structural patterns embedded in India's digital payment ecosystem.

The central thesis of this report is that UPI transaction data is not merely a financial record — it is a behavioural fingerprint of 1.4 billion people. Every salary cycle, every festival, every economic crisis, and every policy decision leaves a traceable mark in the data.

> **Core Finding**
> UPI grew 240,000× in volume over nine years. But more remarkable than the scale is what drove it — not one factor, but the compound effect of crisis adoption (Demonetization, COVID), network expansion (21 to 694 banks), and policy design (Zero MDR), each reinforcing the other.

Section 1 — The Growth Story

1.1 Full Growth Timeline

UPI was launched in April 2016 with minimal adoption — 0.09 million transactions in its first operational month (August 2016). The network spent its first year in obscurity, building quietly with 21 partner banks before a single external event changed everything.

By February 2026, UPI processed 20,394 million transactions in a single month, bringing the ten-year cumulative total to 706,864 million transactions. The growth curve is not a straight line — it is a story told in three acts.

Phase	Period	Key Driver	Volume Range
Act 1: Ignition	2016–2018	Demonetization forced adoption	0.09 Mn → 794 Mn
Act 2: Acceleration	2018–2021	Merchant onboarding, Zero MDR	794 Mn → 4,566 Mn
Act 3: Saturation Signal	2021–2026	Micro-payment democratisation	4,566 Mn → 21,703 Mn

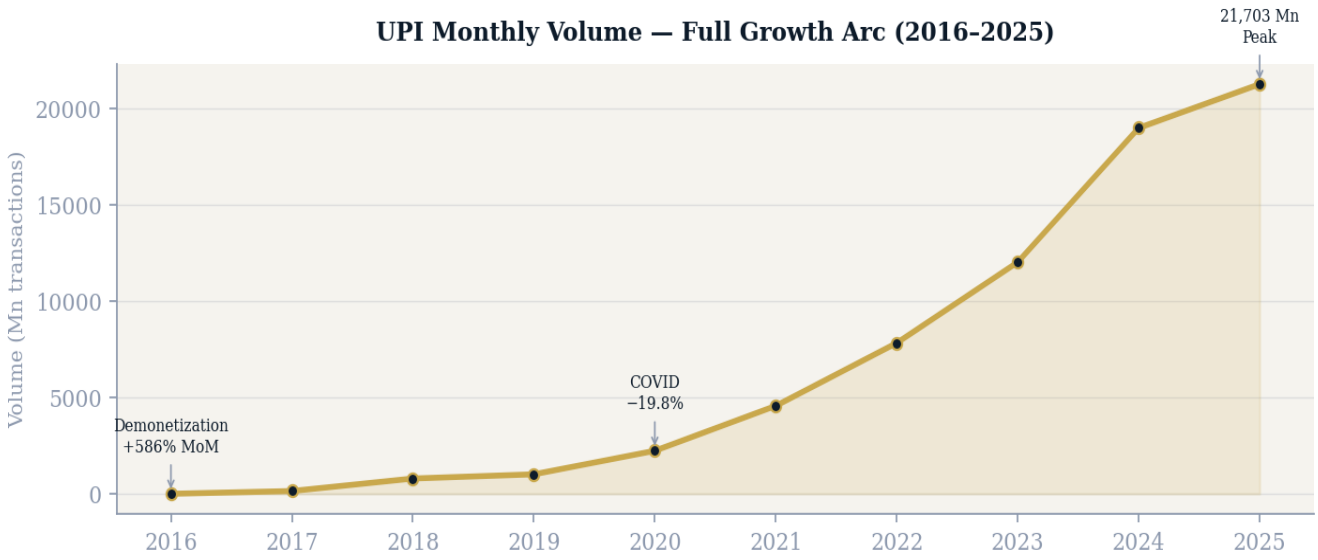

Figure 1: UPI Monthly Volume Growth (Dec-year data points shown) — three-act arc from 1.99 Mn (2016) to 21,261 Mn (2025)

1.2 Demonetization: The Forced Catalyst

On 8 November 2016, the Government of India demonetised ₹500 and ₹1,000 notes — wiping out 86% of currency in circulation overnight. The effect on UPI was immediate and dramatic.

November 2016 saw a 190% month-on-month surge. December 2016 followed with a 586.2% jump — the single largest monthly volume increase in UPI's history. This was not organic growth; it was forced adoption. But forced adoption has a lasting effect: users who discovered UPI during demonetisation never returned to cash at the same rate.

Month	Volume (Mn)	MoM Growth	Banks
Oct 2016	0.10	—	26
Nov 2016	0.29	+190%	30
Dec 2016	1.99	+586%	35
Jan 2017	4.46	+124%	36

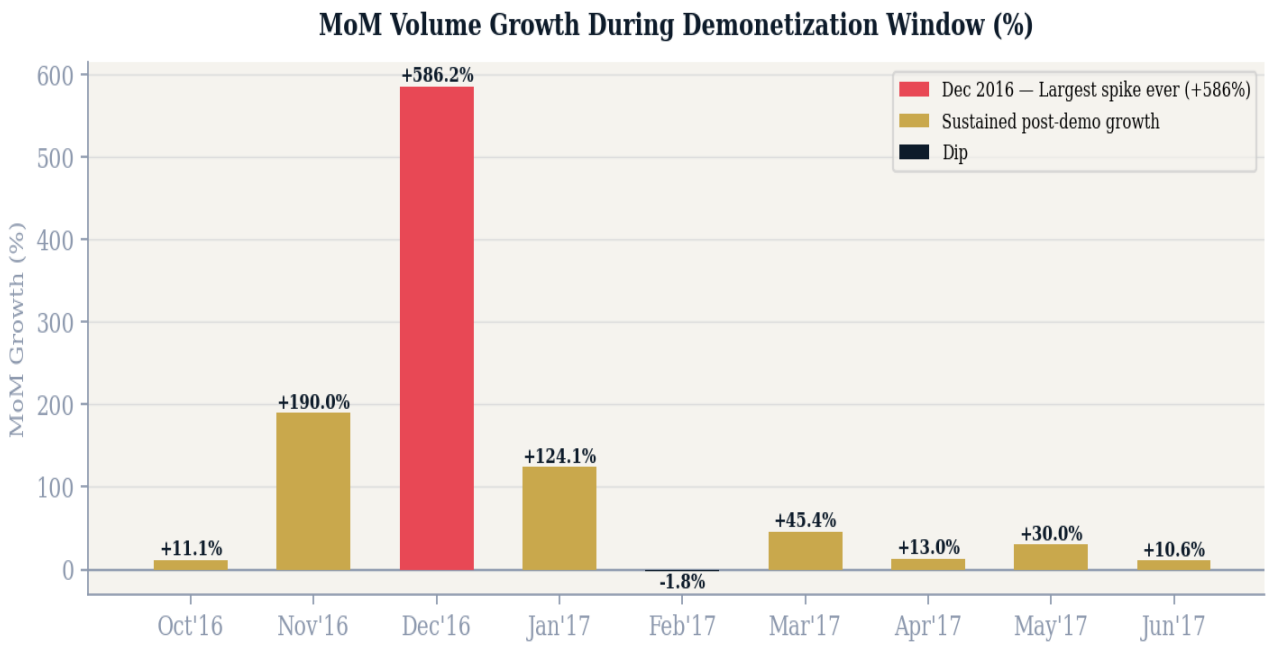

Figure 2: Month-on-Month volume growth (%) during the Demonetization window — Dec 2016 spike of +586% stands unmatched in UPI history

> **Ticket Size Crash**
> As mass users joined during demonetisation, the average ticket size fell from ₹4,857 (October 2016) to ₹917 (October 2017). This is not a negative signal — it reflects the democratisation of digital payments from elite bank transfers to everyday small transactions.

1.3 COVID: The Only Decline — and the Fastest Recovery

April 2020 produced UPI's only sustained volume decline. A national lockdown saw transactions fall 19.8% month-on-month to 999.57 million — the first-time volume had dropped below 1 billion in over a year. People stopped buying. Physical retail was shut. UPI had nowhere to go.

The recovery was textbook V-shaped. May 2020: +23.5%. June 2020: +8.3%. By October 2020 — six months after the collapse — monthly volume had more than doubled to 2,071 million, surpassing all pre-COVID levels. The crisis did not damage UPI; it gave millions more people a reason to use it.

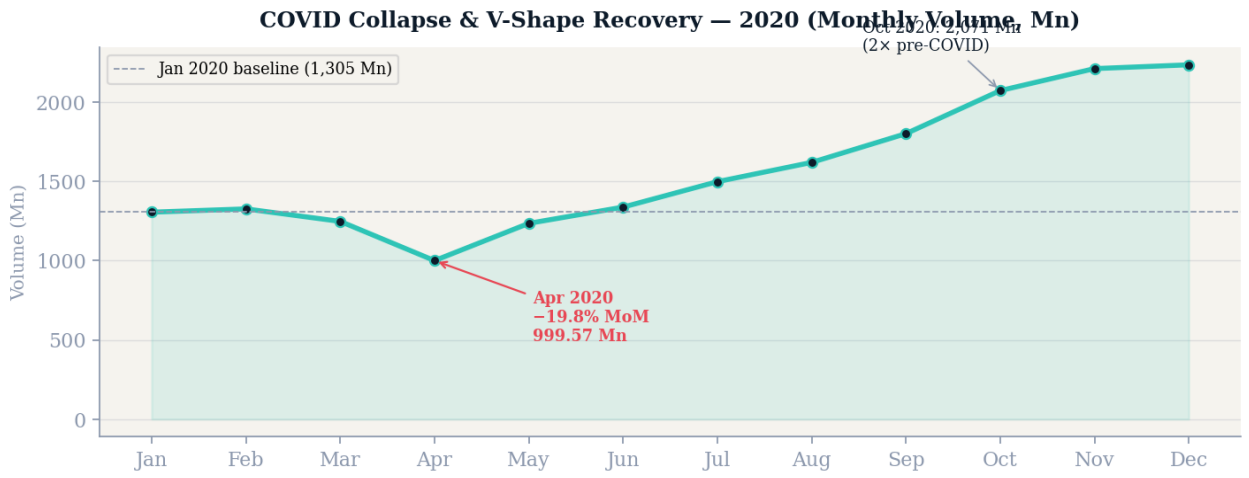

Figure 3: UPI monthly volume during 2020 — the V-shape recovery from April's 999 Mn collapse to October's 2,071 Mn is the sharpest recovery in the dataset

Section 2 — Behavioural Fingerprint

Perhaps the most revealing dimension of this study is what the data says about human behaviour. UPI is not just a payment method — it is a mirror of how Indian households manage money across days, weeks, and months.

2.1 Day of Week: The Character of Transactions

A naive hypothesis would be that weekend usage is lower — people are busy enjoying themselves. The data tells a more nuanced story.

Day	Avg Volume (Mn)	Avg Ticket (₹)	Character
Monday	374.3	₹1,653	B2B settlements, bill payments
Saturday	379.3	₹1,505	Highest volume, leisure spending
Sunday	370.6	₹1,245	Lowest volume AND lowest ticket
Weekday Avg	375.9	₹1,610	—
Weekend Avg	375.0	₹1,375	17% lower ticket than weekday

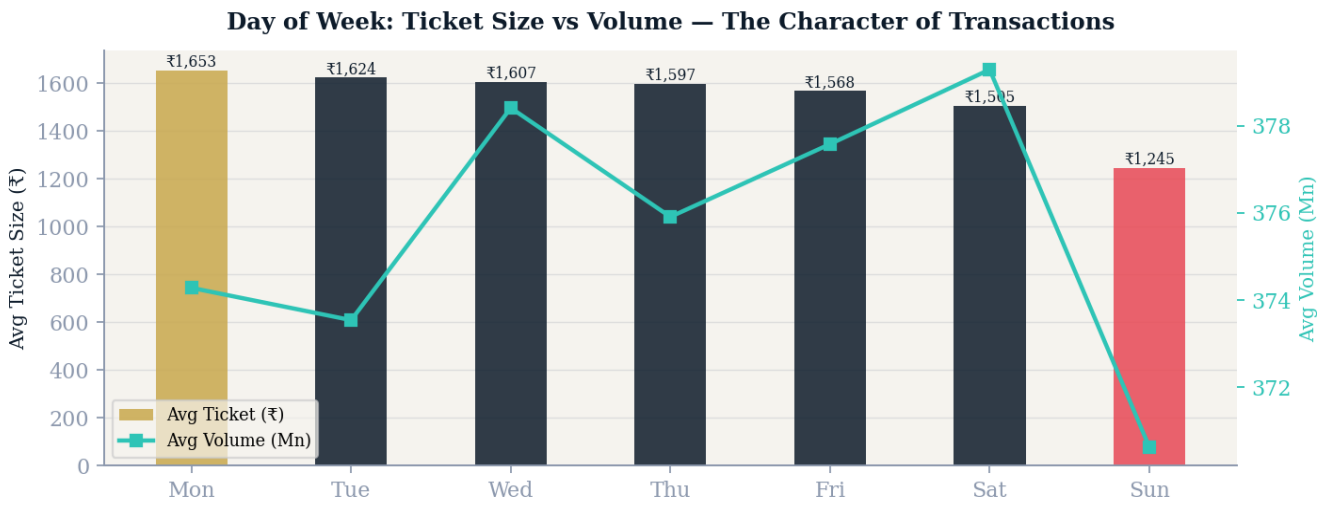

Figure 4: Avg ticket size (bars) and avg volume (line) by day of week — volume is flat, but ticket size drops 33% from Monday to Sunday

Volume is effectively flat across Monday through Saturday — a difference of just 0.3% between weekday and weekend averages. UPI usage frequency is a formed habit. What changes is the purpose: weekday transactions tend to be higher-value business and financial payments; weekend transactions shift toward social, retail, and small-ticket spending.

2.2 Salary Cycle: Three Phases of the Month

The day-of-month analysis reveals a clear three-phase financial calendar embedded in UPI behaviour — a pattern that mirrors how salaried India manages money.

- **FLUSH (Days 1–10):** Rent, EMIs, insurance premiums, and investment SIPs dominate the first week.
- **DRIFT (Days 11–26):** Ticket size steadily declines to a trough of ₹1,405 on Day 26 — everyday retail, food, transport.
- **RECOVERY (Days 28–31):** Ticket size recovers to ₹1,644–1,708 as advance salaries arrive and month-end obligations are settled.

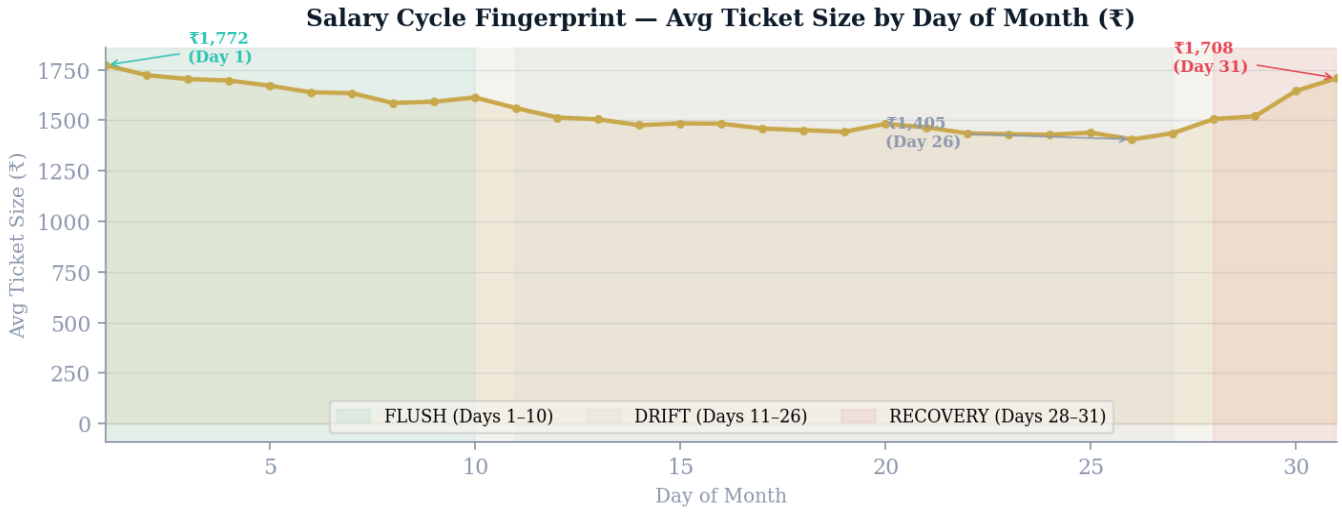

Figure 5: Avg ticket size by day of month — three phases (Flush, Drift, Recovery) visible every month with 21% peak-to-trough swing

> **21% Peak-to-Trough Drop**
> Ticket size falls from ₹1,772 on Day 1 to ₹1,405 on Day 26 — a 21% drop within a single month, every month, as financial pressure normalises into daily spending.

2.3 Festival Paradox

Conventional wisdom holds that festivals drive spending. UPI data challenges this at the transaction level. Across four major festivals — Diwali, Holi, Eid, and Christmas — ticket size falls consistently on the festival day itself, then recovers the next day.

Festival	Before (₹)	Festival Day (₹)	After (₹)	Drop
Diwali	₹1,615	₹1,358	₹1,508	−16%
Eid	₹1,812	₹1,574	₹1,600	−13%
Holi	₹1,460	₹1,279	₹1,527	−12%
Christmas	₹1,421	₹1,296	₹1,452	−9%

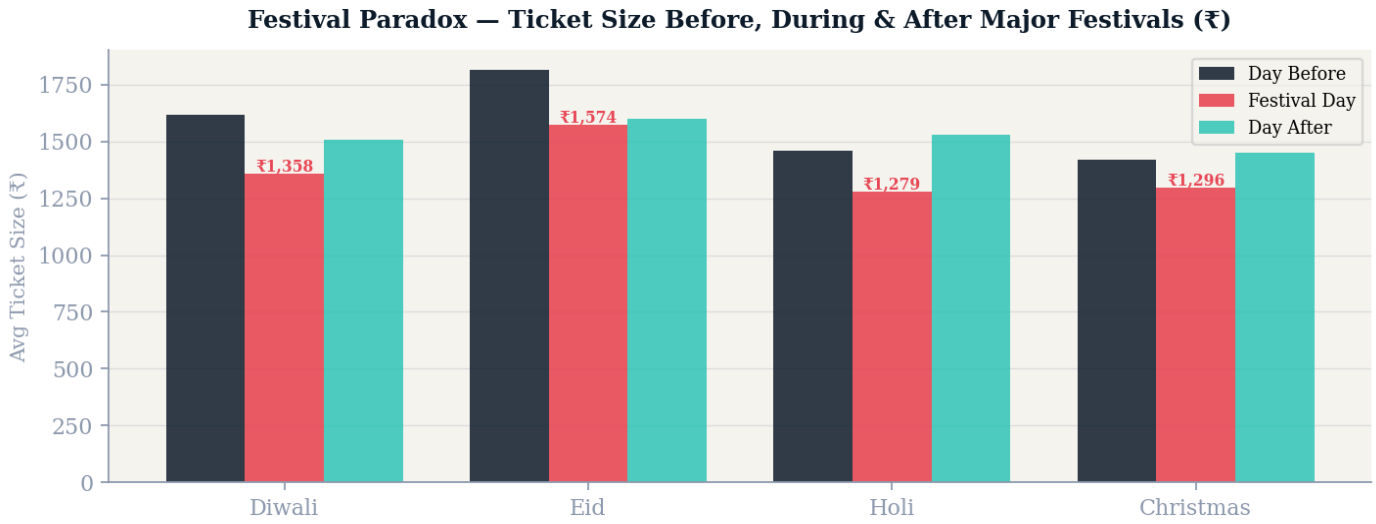

Figure 6: Festival Paradox — ticket size before, on, and after major festivals. Red bars (festival day) consistently the lowest across all four festivals

Volume also dips on festival days. People are not transacting less because they are spending less — they are transacting differently. Celebrations shift payments toward cash gifts, family expenses, and informal exchanges that may not pass through UPI. The day-after recovery confirms that UPI usage is not lost, merely deferred.

Section 3 — The Shrinking Ticket

The most structurally significant finding in this study is the sustained, decade-long decline in average transaction value — from ₹2,648 in FY2015-16 to ₹1,310 in FY2024-25. This is not a failure of UPI. It is the clearest possible signal of financial democratisation.

Fiscal Year	Avg Ticket (₹)	Total Volume (Mn)	Interpretation

FY 2015-16	₹2,648	2.65	Elite early adopters — large bank transfers
FY 2017-18	₹1,518	3,746	Post-demonetisation mass entry; ticket crash
FY 2020-21	₹1,855	38,745	COVID: small payments paused; ticket temporary spike
FY 2022-23	₹1,561	1,17,641	Zero MDR effect: micro-merchants onboarded
FY 2024-25	₹1,310	2,28,282	Structural floor forming; micro-payments dominant

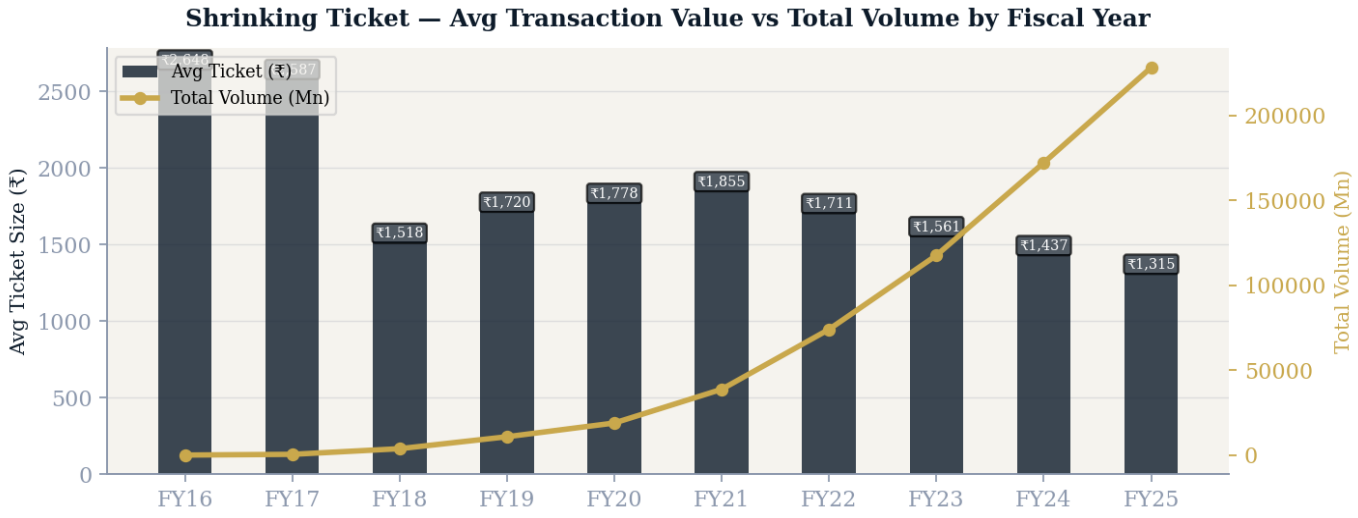

Figure 7: Avg ticket size (bars, left axis) vs total volume (line, right axis) by fiscal year — inverse relationship: as more people join, avg transaction shrinks

3.1 Transaction Size Migration

The daily data reveals a clear migration across ticket size buckets over time:

- **High Value (₹2,000+):** 82 days in this range — all between May 2021 and April 2023. High-value days have effectively vanished.
- **Mid Value (₹1,500–1,999):** 864 days — the largest bucket by count, spanning the full observation period. The core of UPI's value base.
- **Low Value (₹1,000–1,499):** 791 days and growing — increasingly dominant as UPI reaches merchants, kirana shops, and daily commutes.
- **Micro (Below ₹1,000):** 28 days — appearing from August 2023. These are high-volume days (avg 589 Mn transactions) driven by micro-payments below ₹100.

> **Zero MDR Policy Impact**
> The January 2020 elimination of Merchant Discount Rate on UPI transactions removed the last financial barrier for small merchants. The subsequent acceleration of low-value and micro-payment days in the data is the clearest quantifiable evidence of this policy's impact.

Section 4 — Network Effect: Banks as Infrastructure

When UPI launched in August 2016, 21 banks were on the network. By February 2026, 694 banks participated. This 33× growth in network nodes produced a 2,900× increase in volume per bank — one of the clearest demonstrations of Metcalfe's Law in financial infrastructure.

Milestone	Date	Banks	Volume/Bank (Mn)
Network launch	Aug 2016	21	0.004
100 banks milestone	~2018	~100	~8
Volume/bank crosses 20 Mn	Oct 2022	351	20.8
500 banks milestone	2023	559	~21
Current network	Feb 2026	694	29.4

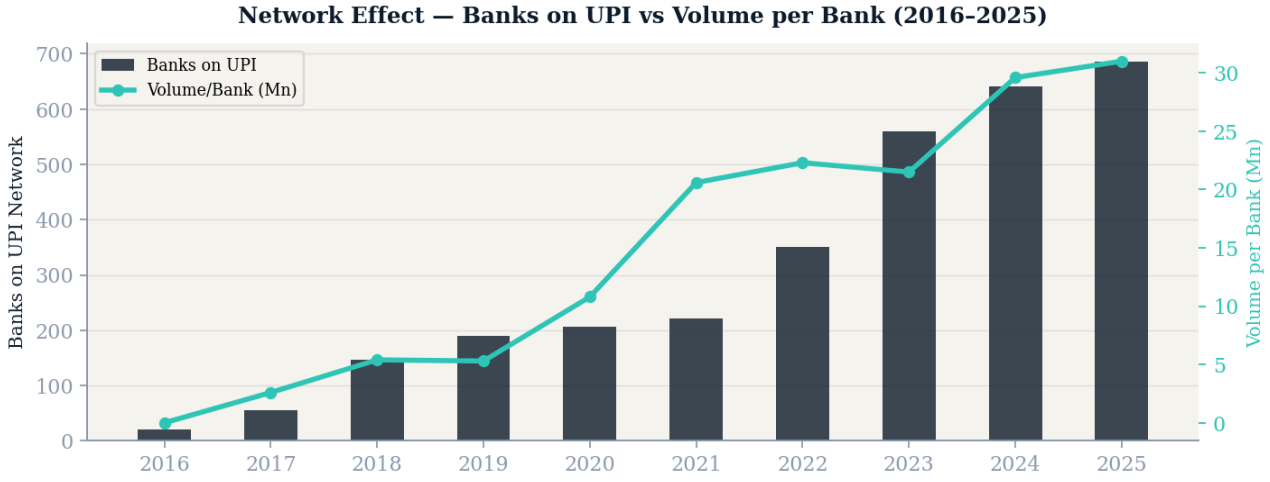

Figure 8: Banks on UPI (bars) vs volume per bank (line) — network effect visible as volume/bank accelerates disproportionately after 2020

The inflection point for volume per bank occurred around 2020–2021, coinciding with COVID-driven adoption and the Zero MDR policy. Beyond a critical network mass (~200 banks, 2019–2020), each new bank added to the UPI ecosystem generated disproportionately more transaction volume — a textbook network effect.

Section 5 — Seasonal Calendar

5.1 Month of Year

A clear W-shaped seasonality pattern emerges when looking at average daily transaction volume by calendar month. February is the surprise leader; May is the consistent laggard.

Month	Avg Daily Volume (Mn)	Rank	Note
February	430.3	#1	Fiscal advance payments + fewer days = highest avg
January	410.0	#2	Year-end financial activity
December	405.0	#3	Year-end surge

May	326.6	#12	Summer slump — lowest volume month
June	335.2	#11	Monsoon onset; slow start

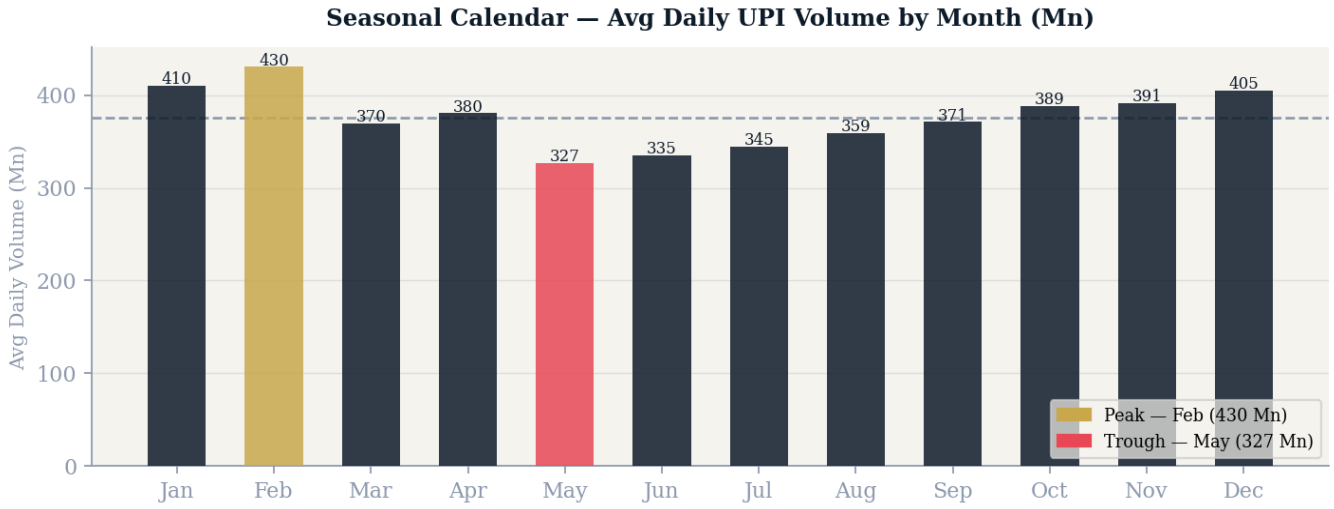

Figure 9: Avg daily UPI volume by calendar month — gold bar (Feb peak) and red bar (May trough) highlight the 24% seasonal swing

5.2 Quarter of Year

The fiscal quarter analysis reveals a counterintuitive result: Q4 (January–March) consistently outperforms Q3 (October–December), despite Q3 containing Diwali and the peak festival season.

Quarter	Avg Daily Volume (Mn)	Avg Ticket (₹)	Interpretation
Q4 (Jan–Mar)	404.9	₹1,531	Tax payments, insurance renewals, advance salary
Q3 (Oct–Dec)	395.0	₹1,534	Festival season — Navratri, Diwali, Christmas
Q2 (Jul–Sep)	358.2	₹1,524	Monsoon; lowest ticket of any quarter
Q1 (Apr–Jun)	344.8	₹1,584	Summer slump; 15% below Q4

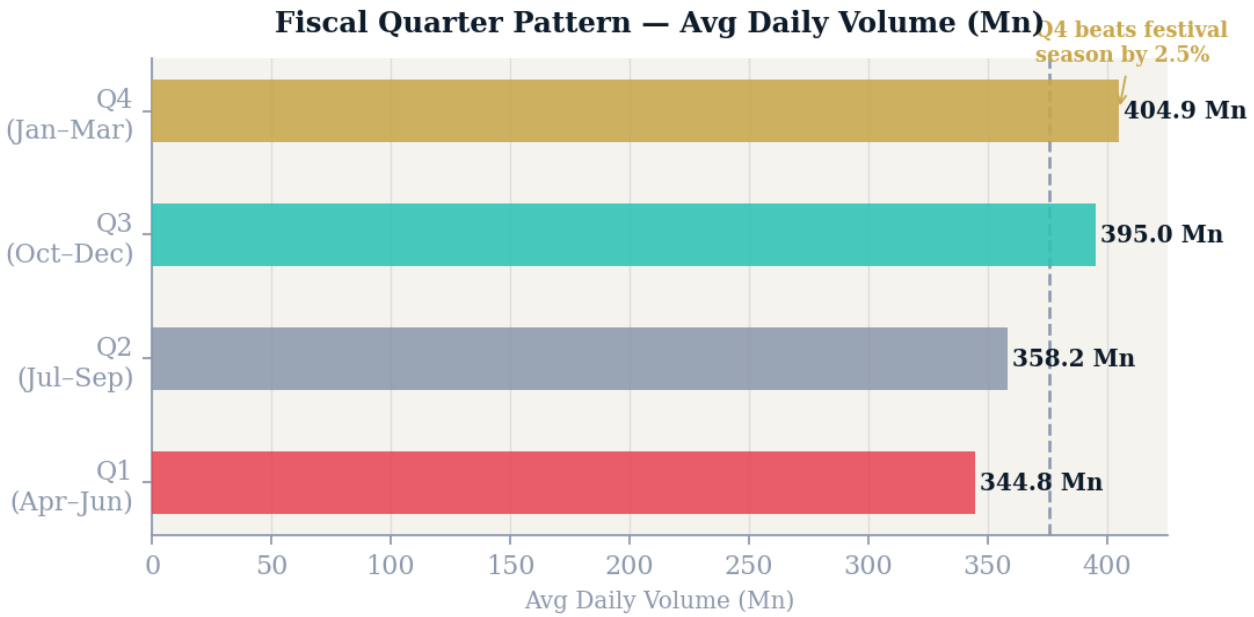

Figure 10: Fiscal quarter ranking by avg daily volume — Q4 (Jan–Mar) tops festival season (Q3) by 2.5%, driven by India's invisible financial calendar

> **The Invisible Financial Festival**
> Q4 is driven by what might be called India's invisible financial festival — the annual ritual of tax filing, advance tax payments, insurance premium renewals, and mutual fund SIPs that concentrates high-value financial transactions in January–March every year, without a single holiday to mark it.

Conclusions & Implications

This analysis supports seven core conclusions about UPI and India's digital payments ecosystem:

- **1. Crisis is the most effective adoption driver.** External shocks — demonetisation, COVID, policy changes — create permanent step-changes in adoption. The floor never goes back to where it was.
- **2. UPI frequency is now a reflex, not a choice.** Volume barely varies by day of the week. The habit of digital payment is fully formed in urban India.
- **3. Financial behaviour is predictable and cyclical.** The three-phase salary cycle (Flush → Drift → Recovery) is visible in transaction data — a 21% peak-to-trough swing, every month.
- **4. Festivals pause high-value transfers.** Ticket drops 9–16% on major festival days across all religions. Celebrations prioritise presence over payment.
- **5. Smaller ticket = more people included.** The ₹2,648 → ₹1,310 ticket decline represents success, not failure. UPI is now used to buy chai, not just to transfer salaries.
- **6. UPI is infrastructure, not just a product.** Each new bank on the UPI network generates disproportionate transaction volume — a textbook demonstration of network effects.
- **7. Q4 is the biggest quarter — and it has no holiday.** India's year-end financial obligations generate more UPI volume than even the festival season.

The broader implication is this: UPI is no longer a technology story. It is a behavioural economics story. The patterns in this data are the patterns of how a society manages money — its fears, its rhythms, its celebrations, and its obligations. Understanding them has practical value for policymakers, fintech product teams, and anyone trying to serve the next billion digital payment users.

Appendix — SQL Queries Overview

Query ID	Section	Key Technique	Primary Finding
Q1.1	Growth Timeline	SUM () OVER window function	0.09 Mn → 21,703 Mn over 9 years
Q1.2	COVID Recovery	LAG () for MoM growth	Only sustained decline: −19.8% Apr 2020
Q1.3	Demonetization	LAG () with NULLIF ()	+586% Dec 2016; ticket crashed ₹4,857 → ₹917
Q2.1	Day of Week	GROUP BY Day_of_Week	33% ticket gap Mon (₹1,653) vs Sun (₹1,245)
Q2.2	Weekend Split	Is_Weekend flag grouping	Volume flat; value gap 14.3%; ticket gap 17%
Q2.3	Salary Cycle	GROUP BY Day_of_Month	₹1,772 Day 1 → ₹1,405 Day 26 → ₹1,708 Day 31
Q2.4	Seasonality	DATENAME(MONTH)	Feb highest (430 Mn); May lowest (327 Mn)
Q2.5	Festival Paradox	CASE WHEN date IN (...)	Diwali day ticket −16% vs day before
Q3.1	Annual Ticket	Fiscal year CASE grouping	₹2,648 (FY16) → ₹1,310 (FY25)
Q3.2	Monthly Ticket	Month_Year aggregation	Aug 2025 all-time low: ₹1,239
Q3.3	Segmentation	CASE ticket bucket	High-value days ended Apr 2023; Micro from Aug 2023
Q4.1	Bank Network	Volume / Banks_On_UPI	2,900× volume/bank growth (0.01 → 29.4 Mn)
Q5.1	Quarterly	CASE WHEN MONTH () IN	Q4 > Q3: 404.9 Mn vs 395.0 Mn avg daily

— End of Report —

Data Source: NPCI | Analysis by Pinaki | Tools: SQL Server · Power BI